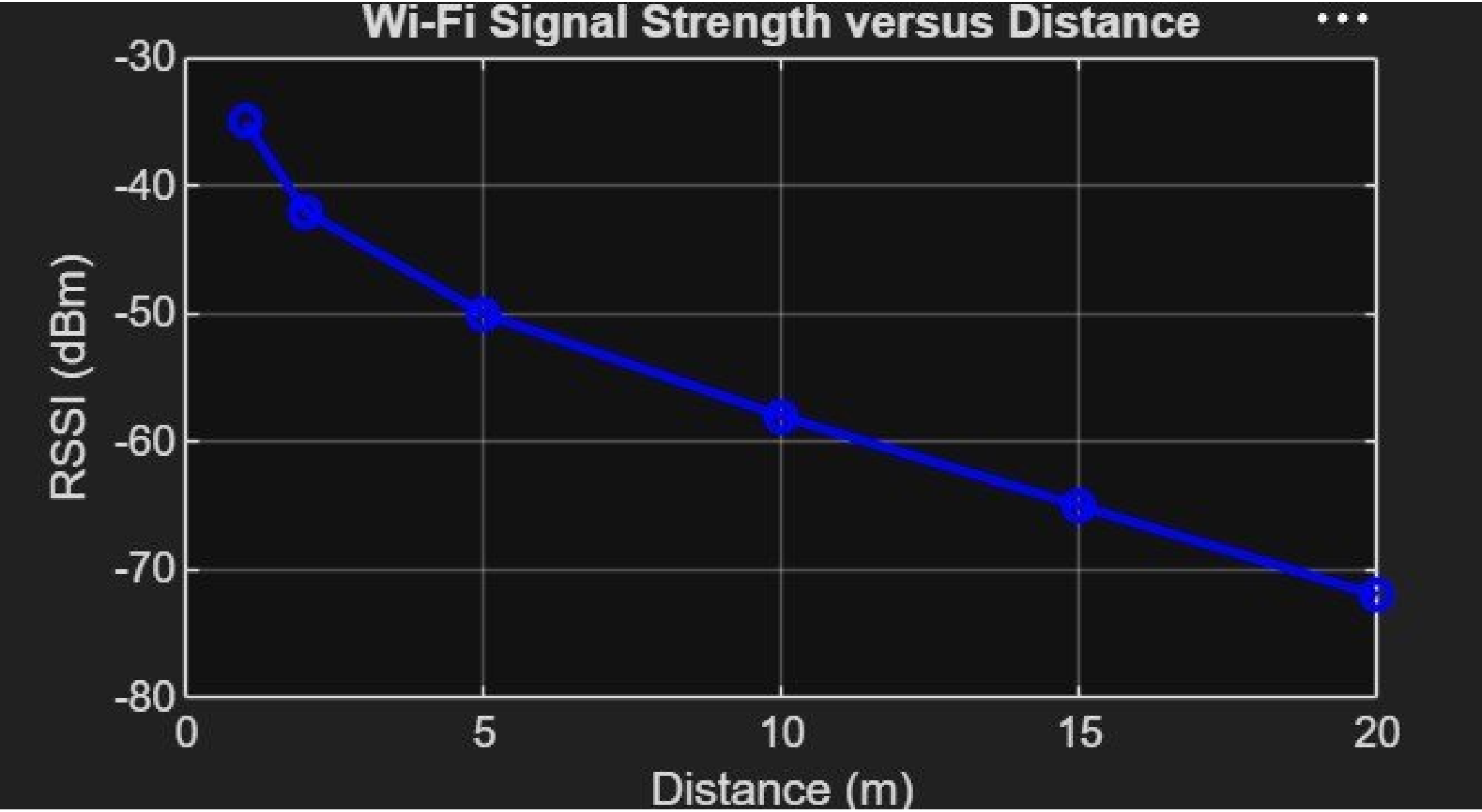
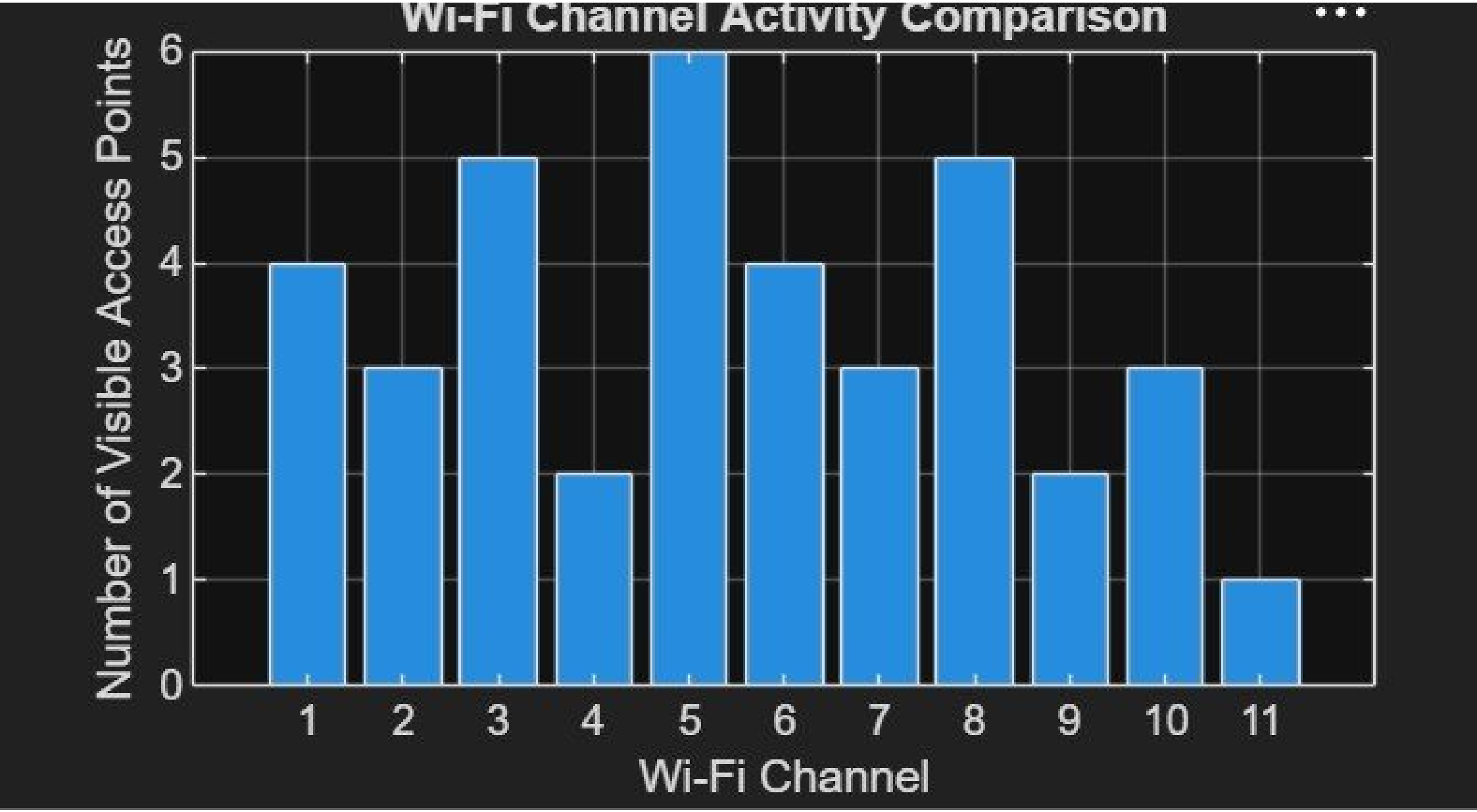
```
clear;
clc;
close all;
% Distance in metres
distance = [1 2 5 10 15 20];
% Illustrative RSSI values in dBm
RSSI = [-35 -42 -50 -58 -65 -72];
% Display results
T = table(distance',RSSI', ...
    'VariableNames',{'Distance_m','RSSI_dBm'});
disp(T);
% Plot RSSI versus distance
figure;
plot(distance,RSSI,'bo-','LineWidth',2);
xlabel('Distance (m)');
ylabel('RSSI (dBm)');
title('Wi-Fi Signal Strength versus Distance');
grid on;
% Illustrative Wi-Fi channel activity
channel = 1:11;
accessPoints = [4 3 5 2 6 4 3 5 2 3 1];
% Display channel activity
T2 = table(channel',accessPoints', ...
    'VariableNames',{'WiFi_Channel','Visible_AccessPoints'});
disp(T2);
% Plot channel activity
figure;
bar(channel,accessPoints);
xlabel('Wi-Fi Channel');
ylabel('Number of Visible Access Points');
```

```
% Display results
T = table(distance',RSSI', ...
    'VariableNames',{'Distance_m','RSSI_dBm'});
disp(T);
% Plot RSSI versus distance
figure;
plot(distance,RSSI,'bo-','LineWidth',2);
xlabel('Distance (m)');
ylabel('RSSI (dBm)');
title('Wi-Fi Signal Strength versus Distance');
grid on;
% Illustrative Wi-Fi channel activity
channel = 1:11;
accessPoints = [4 3 5 2 6 4 3 5 2 3 1];
% Display channel activity
T2 = table(channel',accessPoints', ...
    'VariableNames',{'WiFi_Channel','Visible_AccessPoints'});
disp(T2);
% Plot channel activity
figure;
bar(channel,accessPoints);
xlabel('Wi-Fi Channel');
ylabel('Number of Visible Access Points');
title('Wi-Fi Channel Activity Comparison');
grid on;
% Find channel with minimum visible access points
[~,bestIndex] = min(accessPoints);
fprintf('Channel with minimum visible access points: %d\n', ...
    channel(bestIndex));
```


Wi-Fi Signal Strength versus Distance
-30
-40
-50
-60
-70
-80
RSSI (dBm)
0
5
10
15
20
Distance (m)


Wi-Fi Channel Activity Comparison
Number of Visible Access Points
0
1
2
3
4
5
6
1
2
3
4
5
6
7
8
9
10
11
Wi-Fi Channel

Command Window

```
    WiFi_Channel    Visible_AccessPoints
    ____________    ____________________

           1                    4
           2                    3
           3                    5
           4                    2
           5                    6
           6                    4
           7                    3
           8                    5
           9                    2
          10                    3
          11                    1

Channel with minimum visible access points: 11
>>
           5                    6
           6                    4
           7                    3
```

Distance_m	RSSI_dBm
1	-35
2	-42
5	-50
10	-58
15	-65
20	-72

WiFi_Channel	Visible_AccessPoints
1	4
2	3
3	5
4	2
5	6
6	4
7	3
8	5
9	2